Table I
Performance comparison on Chengdu T5 inflow (32 × 32 grid).

	H3		H6		H12		AVG12	
Model	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE
HA	1.053	1.522	1.083	1.600	1.144	1.739	1.088	1.609
SVR	0.810	1.354	0.835	1.391	0.892	1.499	0.840	1.405
DCRNN (ICLR'18)	0.725	1.321	0.790	1.363	1.025	1.574	0.829	1.399
STGCN (IJCAI'18)	0.721	1.342	0.724	1.348	0.743	1.409	0.728	1.362
GWNet (IJCAI'19)	0.719	1.332	0.731	1.373	0.756	1.450	0.733	1.378
DSTAGNN (ICML'22)	1.022	1.461	1.008	1.458	1.044	1.505	1.027	1.473
PDFormer (AAAI'23)	1.703	2.728	1.704	2.728	1.703	2.727	1.704	2.727
PatchTST (ICLR'23)	0.799	1.405	0.826	1.483	0.880	1.614	0.831	1.490
TimeMixer (ICLR'24)	0.818	1.434	0.844	1.505	0.895	1.620	0.848	1.510
STDN (AAAI'25)	1.178	1.589	1.180	1.593	1.179	1.600	1.178	1.593
MetaDG (AAAI'26)	1.102	1.564	1.170	1.678	1.291	1.882	1.178	1.692
HyperD (AAAI'26)	0.723	1.369	0.723	1.369	0.726	1.382	0.723	1.369
RAST (AAAI'26)	0.724	1.332	0.734	1.372	0.756	1.446	0.736	1.377
CMSRN (Ours)	**0.706**	**1.287**	**0.710**	**1.298**	**0.722**	**1.329**	**0.711**	**1.302**

Table II
Performance comparison on Chengdu T5 outflow (32 × 32 grid).

	H3		H6		H12		AVG12	
Model	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE
HA	1.023	1.404	1.043	1.439	1.085	1.510	1.047	1.445
SVR	0.792	1.239	0.809	1.251	0.838	1.285	0.810	1.255
DCRNN (ICLR'18)	0.714	1.230	0.761	1.237	1.151	1.642	0.844	1.326
STGCN (IJCAI'18)	0.714	1.235	0.716	1.239	0.725	1.258	0.718	1.242
GWNet (IJCAI'19)	0.713	1.253	0.718	1.265	0.729	1.289	0.719	1.267
DSTAGNN (ICML'22)	1.018	1.383	1.007	1.376	1.035	1.410	1.021	1.390
PDFormer (AAAI'23)	1.601	2.375	1.601	2.375	1.602	2.376	1.601	2.375
PatchTST (ICLR'23)	0.786	1.300	0.802	1.333	0.837	1.396	0.805	1.337
TimeMixer (ICLR'24)	0.807	1.315	0.823	1.345	0.856	1.404	0.826	1.349
STDN (AAAI'25)	1.135	1.501	1.135	1.501	1.135	1.503	1.134	1.500
MetaDG (AAAI'26)	1.082	1.471	1.139	1.543	1.245	1.672	1.146	1.550
HyperD (AAAI'26)	0.712	1.249	0.712	1.249	**0.714**	1.254	0.712	1.249
RAST (AAAI'26)	0.718	1.242	0.724	1.257	0.735	1.282	0.724	1.258
CMSRN (Ours)	**0.708**	**1.223**	**0.710**	**1.226**	0.718	**1.239**	**0.711**	**1.228**

Table III
Performance comparison on Shanghai T5 inflow (32 × 32 grid).

	H3		H6		H12		AVG12	
Model	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE
HA	1.019	1.437	1.040	1.490	1.080	1.593	1.043	1.498
SVR	0.817	1.251	0.837	1.274	0.879	1.344	0.841	1.283
DCRNN (ICLR'18)	0.734	1.243	0.748	1.262	0.808	1.328	0.758	1.272
STGCN (IJCAI'18)	0.741	1.257	0.743	1.260	0.754	1.293	0.745	1.268
GWNet (IJCAI'19)	0.735	1.268	0.742	1.288	0.760	1.343	0.744	1.295
DSTAGNN (ICML'22)	1.010	1.410	0.950	1.360	1.009	1.426	0.994	1.402
PDFormer (AAAI'23)	1.592	2.472	1.591	2.471	1.589	2.468	1.591	2.471
PatchTST (ICLR'23)	0.800	1.333	0.816	1.382	0.849	1.480	0.819	1.392
TimeMixer (ICLR'24)	0.821	1.331	0.836	1.372	0.868	1.463	0.839	1.381
STDN (AAAI'25)	1.219	1.651	1.220	1.659	1.219	1.683	1.218	1.661
MetaDG (AAAI'26)	1.082	1.517	1.168	1.624	1.421	1.944	1.201	1.667
HyperD (AAAI'26)	0.737	1.261	0.737	1.261	0.737	1.263	0.737	1.260
RAST (AAAI'26)	0.736	1.245	0.740	1.259	0.749	1.289	0.741	1.261
CMSRN (Ours)	**0.726**	**1.221**	**0.727**	**1.225**	**0.734**	**1.241**	**0.728**	**1.228**

Table IV
Performance comparison on Shanghai T5 outflow (32 × 32 grid).

	H3		H6		H12		AVG12	
Model	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE
HA	1.059	1.528	1.079	1.559	1.118	1.629	1.082	1.565
SVR	0.866	1.373	0.882	1.395	0.914	1.447	0.883	1.398
DCRNN (ICLR'18)	0.782	1.355	0.794	1.365	0.856	1.388	0.804	1.365
STGCN (IJCAI'18)	0.784	1.353	0.787	1.359	0.797	1.385	0.789	1.364
GWNet (IJCAI'19)	0.783	1.381	0.788	1.394	0.801	1.424	0.789	1.396
DSTAGNN (ICML'22)	1.017	1.462	0.994	1.446	1.080	1.535	1.016	1.469
PDFormer (AAAI'23)	1.720	2.726	1.720	2.726	1.719	2.724	1.720	2.726
PatchTST (ICLR'23)	0.848	1.439	0.863	1.465	0.897	1.533	0.866	1.473
TimeMixer (ICLR'24)	0.869	1.448	0.883	1.473	0.915	1.538	0.886	1.480
STDN (AAAI'25)	1.109	1.591	1.107	1.593	1.105	1.605	1.106	1.595
MetaDG (AAAI'26)	1.127	1.600	1.245	1.748	1.599	2.238	1.298	1.825
HyperD (AAAI'26)	0.778	**1.344**	0.778	**1.344**	**0.778**	**1.347**	0.778	**1.344**
RAST (AAAI'26)	0.787	1.366	0.792	1.380	0.804	1.409	0.793	1.381
CMSRN (Ours)	**0.773**	1.346	**0.774**	1.345	0.778	1.353	**0.774**	1.346